

Falgun Tanwar

R&D (Machine Learning & IoT Mobile Technologies),
Cross platform mobile app development- Flutter

Location: Jaipur, India
falguntnwr3@gmail.com

Portfolio:
<https://wisdomg1.github.io/portfolio>

EXPERIENCE

Teck Valley India Pvt Ltd (Virtual Employee)
Sr. Software Engineer (Flutter Development)
01/2022 - Present

Tweaking Technologies
Android Developer
05/2019 - 01/2022

Systweak Software
Android Developer
05/2015 - 08/2017

Pegasus Technologies
Sr. Android Developer
09/2013 - 04/2015

RRSC/ISRO –Jodhpur
Android Trainee
2012-2013

Samarth BCA college
Lecturer (C Language, Network Technologies)
(2010-2011)

EDUCATION

Master of Technology (Remote Sensing & GIS),
Banasthali University,
Niwai campus, Rajasthan, India
2011 – 2013
Integrated Master of Science (Information Technology),
Gujarat University,
Ahmedabad, Gujarat, India
2005 – 2010

PROJECTS

Flutter:

FUTURA Business Software: [link for apple store](#), [link for google play store](#)
An AI-enabled utility app for companies and their clients to smartly scan bills and invoices and manage business reports

Role: Team Member
Responsibility: design, project architecture and development for mobile app

PROFILE

Cross Platform App development with Flutter-Dart

Machine Learning based TFlite Model writing in Google Colab and integration in mobile app

Android based App development with Java.

SKILLS

IDE: Android Studio, Google Colab, Visual Studio, IBM Watson (beginner)

Programming: Core Java, Dart, And Python.

Project Management Tool: JIRA, Zoho Projects, slack, trello board

WEB API: REST API calling, JSON Handling, Signal R, WebSocket,

Services: Background Task and background services, Firebase services, Firebase ML kit

Google Ads/Promotional Ads: google Ads SDK integration and Custom Library making for cross product promotion.

ML Libraries & Tools: OpenCV, Google Teachable Machines, keras open data sets, Tensorflow, Scikit-Learn, NumPy, SciPy, Pandas, NLTK, Matplotlib, seaborn.

CERTIFICATIONS

Artificial Intelligence Engineer Master Program
Simplilearn | 10/2019 – 12/2020

ADDITIONAL CERTIFICATIONS

Certificates of participation in Various National/ International Symposium, workshops and seminars and Webinars in the field of Remote Sensing and GIS, Advance Computing Techniques, open source Technologies, AI and Robotics, Sponsored by ISRO, DRDO, IIRS, IGRS etc.

Team Size: 8

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, AI Apis, Document scanning.

NewYork Celebrity Guide:

An app to know more about your favorite celebrity from new york. [link for apple store](#), [link for google play store](#)

Role: Project Lead

Responsibility: Project architecture, team management

Team Size: 3

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, SQLite Database, Firestore, Firebase analytics and Google Tag Manager for usage tracking.

Paris Celebrity Guide:

An app to know more about your favorite celebrity from paris. [link for apple store](#), [link for google play store](#)

Role: Project Lead

Responsibility: Project architecture, team management

Team Size: 3

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, SQLite Database, Firestore, Firebase analytics and Google Tag Manager for usage tracking.

Brain Game: [url](#)

an effective app for brain exercise.

Role: Project Lead

Responsibility: Design, project architecture and development, team management

Team Size: 3

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, SQLite Database, Firestore.

Android Java:

PAC-BLU NexGen [url](#)

An IoT enabled Utility app to manage office floor door lock /unlock based for assigned employees/users.

Responsibility: design, project architecture and development.

Team Size: 4

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, GETX, BLE Communication with bluetooth devices, IOT, Github.

Blue Star Driver [url](#)

An IoT enabled Utility app to manage truck driving for drivetime, route logs, halts and duty time, logs and penalties.

Responsibility: design, project architecture and development, team management.

Team Size: 4

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin, Dart programming, GETX, BLE Communication with bluetooth devices, IOT, Github.

Systweak ABCD Drawing App: [url](#)

A handy tool for toddlers to learn alphabet writing using tracing the letters and numbers.

Role: Project Lead,

Certificates of participation in different coding challenges By Hacker Rank Techgig,

ACHIEVEMENTS

Techgig Android Skill Analysis rank 370

Secure the 1st position in poster Present on “Review on Geospatial Technology Solutions for Seismic Hazards Assessment” At IGRS Symposium.

The paper on “Review on Geospatial Technology Solutions for Seismic Hazards Assessment” was selected for ACTA Geo-Physica peer review.

Secured First Position In “English Writing skills” in class X.

“GARGI” Award for Class X performance.

LANGUAGES

English, Hindi, Gujarati

Responsibility: design, architecture and development with team handling.

Team Size: 2

Development Tool and Tech: Android Studio, Java programming, Android Canvas path drawing, path matching Bitmap pixel manipulation.

Photos Recovery: [url](#)

Utility app for recovering accidentally deleted pictures.

Role: Project Lead

Responsibility: design, architecture and development and testing and support.

Team Size: Individual

Development Tool and Tech: Android Studio, Java programming, Background Task for reading storage, Image Loader, file saving, Permission handling. Google Ads sdk integration

Locker For Insta Social App: [url](#)

A necessary tool to lock the Instagram Direct user chat.

Role: Project Lead,

Responsibility: design, architecture and development and testing and support.

Team Size: Individual

Development Tool and Tech: Android Studio, Java programming, Accessibility Service. Window Manager remote View lock. Google Ads sdk integration

Check Data Usage: [url](#)

Utility app for knowing mobile data and Wi-Fi data usage..

Role: Project Lead,

Responsibility: design, architecture and development and testing and support.

Team Size: Individual

Development Tool and Tech: Android Studio, Java programming, native function for getting mobile data and wifi usage. Speed test API. Google Ads sdk integration

Duplicate File Fixer: [url](#)

utility app for finding all duplicate media files

Role: Project module design and development,

Responsibility: Design and development and testing and support for the module.

Team Size: Individual

Development Tool and Tech: Android Studio, Java programming, Comparing the files with their size, type and content hash code.

Others most popular app on store:

PhotosCleaner, Medicine Reminder, and Social Fever

Enterprise Edition:

Phomello POS System for Restaurant. [url](#)

Remote Employee attendance system.

R&D/MVP/ Prototype Apps

Lock Cart:

A utility app for supermarket to manage luggage trolley

Role: Project Lead

Responsibility: design, project architecture and development, team management.

Team Size: 4

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin,Dart programming, GETX, BLE Communication with bluetooth devices, IOT.

Pneumo Controller

A utility app for reading continuous health data(United State User) from the bluetooth enabled pneumo devices.

Role: IoT Project Manager

Responsibility: design, project architecture and development, project management.

Team Size: 2-6

Development Tool and Tech: Github, Zoho Projects, Cross Platform, Android Studio, Flutter Plugin,Dart programming, GETX, BLE Communication with bluetooth devices, IOT.IBM watson, Arduino.

Pneumo Controller:(The Admin Web Portal)

A display portal to view the user health data for admin and user.

Responsibility: design, portal architecture and development, team management.

Team Size: 3

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin,Dart programming, GETX, BLE Communication with bluetooth devices, IOT.Zoho Projects, github.

Smart Garden App

An interesting utility app for controlling & managing the smart planter for managing plant life cycle from start to end and watering.

Responsibility: design, project architecture and development, team management.

Team Size: 4

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin,Dart programming, GETX, BLE Communication with bluetooth devices, IOT,Github.

Canasta Game:

4 player based Online card game

Role: Project Lead

Responsibility: design, project architecture and development, team management.

Team Size: 4

Development Tool and Tech: Cross Platform, Android Studio, Flutter Plugin,Dart programming, Firebase Authentication (Google,OTP), SQLite Database,AWS WebSocket communication for Server DB Queries.

Case Studies:

AI Master Program Case Studies

Tools and Technology used: Google Colab, python, csv files, Pandas data frames, Matplotlib, nltk corpus, stopwords, spacy, gensim, pyLDAvis

- Data Science Project: Movie lens Case Study to predict the top movies based on user ratings. • Include: merging data frame, Exploratory Data analysis and data visualization.
- Machine Learning Project: Income Qualification Case Study to identify the level of income qualification needed for the families in Latin America • Include: data cleaning, EDA.
- Deep Learning Project: Pet Classification Case Study: Classify the given pet image to the specific category (cat or dog) • Include: image prepossessing, Image data generator. Sequential model with multiple layers. Batch normalization, relu and softmax activation function.
- Natural Learning Project: Topic Analysis of review data: find the most frequent words used in user reviews. Include: stop words, lemmatization, tokenization, visualization of most common words, LDA Algo, topic visualization. AI Capstone Project: Sales prediction for retail store: complete 4 week task (EDA, Different ML, DL, model and compare for better result) • Include: EDA, Time Series, LSTM , Random forest, XGboost, RMSE,MAE.

TFLite Model Integration in flutter/ java apps:

Tools and Technology used: Google Colab, python, Google Teachable Machines, Keras open datasets.Open CV Library

- Object recognition using Firebase ML Kit
- cat vs dog classification in android/Flutter
- Mask Detection of camera images in android/Flutter

Master program (GIS and Remote Sensing) Case Studies

Tools and Technology used: Eclipse, Android, Java, Open Street maps,ShapeFiles, SQLite

Mobile GIS based Agriculture Application for farmers,

- to collect soil sample,drawing farm boundary on the map (overlays)
- SQLite Database , storage of data.